CHAPTER 10

Graphs

Vocabulary, degrees, adjacency matrix, counting walks

Objectives. Model a situation with a graph; use the vocabulary of graphs (order, degree, walk, cycle, connectedness, complete graph, directed graph); write the adjacency matrix of a graph; count the walks of given length with the powers of this matrix.

Prerequisites. Chapter 9 (product and powers of matrices); proof by induction; counting.

Fluency

Fluency · No calculator, five minutes.

1. Compute $\binom{4}{2}$ and $\binom{8}{2}$.
2. Compute $1 + 2 + \cdots + 7$.
3. Compute $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^2$.
4. In a group of 5 people, each shakes hands with all the others. How many handshakes are there?
5. A sum of numbers is odd. What can be said about the number of odd terms?
6. Write the entry in row i and column j of M^2 in terms of the entries of M, for M of order n.